

CHARLOTTESVILLE 2 SSE, VA, USA

WMO: 746010

Lat: 37.9975N

Lon: 78.4656W

Elev: 1177

StdP: 14.08

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 03759

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.3	17.4	-6.2	4.1	15.0	-1.7	5.2	19.9	12.6	48.7	11.3	47.4	3.6	N/A	0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	89.1	73.5	86.7	72.5	84.5	71.6	75.5	85.0	74.3	83.4	73.2	81.6	3.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.7	126.8	80.7	71.4	121.1	78.8	70.4	117.0	77.8	39.8	85.3	38.6	83.5	37.5	82.0	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	8.7	7.8	8.6	92.3	5.1	2.9	4.9	94.4	2.0	96.1	-0.9	97.8	-4.5	99.9		
			5.2	77.5	5.0	0.8	1.6	78.1	-1.4	78.5	-4.2	79.0	-7.8	79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	34.8	38.9	45.9	56.5	65.2	72.5	75.9	74.1	68.8	58.8	47.0	40.3	
	DBStd	16.42	10.12	10.83	10.61	9.35	7.56	5.32	4.83	4.61	6.57	8.44	8.70	8.96	
	HDD50	1566	480	337	211	40	1	0	0	0	0	19	158	320	
	HDD65	4227	936	730	596	284	97	8	0	2	36	231	542	765	
	CDD50	3993	9	27	83	235	471	674	804	749	563	291	66	21	
	CDD65	1179	0	0	3	29	102	232	340	285	149	38	1	0	
	CDH74	7697	0	0	14	182	527	1487	2616	1842	855	170	4	0	
	CDH80	1949	0	0	0	22	75	350	802	463	205	32	0	0	
Wind	WSAvg	3.8	4.4	4.6	4.4	4.6	3.6	3.2	3.0	2.9	3.0	3.8	4.0	4.2	
Precipitation	PrecAvg	47.5	3.3	3.2	3.9	3.2	4.5	4.0	5.2	4.3	4.6	4.1	3.6	3.3	
	PrecMax	74.6	9.7	8.0	8.3	9.8	10.6	12.8	14.8	10.5	18.0	12.7	13.1	7.4	
	PrecMin	30.4	0.2	0.2	0.2	0.9	1.5	0.4	1.1	1.0	0.6	0.0	0.6	0.4	
	PrecStd	9.3	1.9	1.9	1.8	1.8	2.0	2.8	3.0	2.1	3.9	3.0	2.5	1.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.0	69.9	76.1	83.1	85.2	90.2	93.2	90.4	89.1	83.9	72.8	67.4	
		MCWB	54.8	58.5	57.9	66.4	69.1	74.4	75.1	71.9	71.6	69.8	60.8	60.0	
	2%	DB	59.1	64.4	70.9	79.1	82.2	86.7	90.0	87.2	85.3	78.5	66.4	62.6	
		MCWB	51.7	54.1	56.4	62.3	68.6	72.2	74.2	73.5	70.5	67.0	56.2	57.1	
	5%	DB	55.0	60.2	66.9	75.4	79.7	84.1	87.1	84.9	82.2	74.7	63.4	58.6	
		MCWB	47.5	51.2	53.9	61.1	68.1	71.3	73.3	72.3	69.8	65.2	53.5	52.6	
	10%	DB	49.8	55.6	62.8	71.4	76.9	81.5	84.5	82.4	79.1	71.3	60.4	54.2	
		MCWB	42.7	46.6	51.8	58.4	66.0	69.8	72.5	71.1	68.2	63.1	51.3	47.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.5	60.4	62.3	68.8	73.1	75.5	77.7	76.4	74.4	71.5	62.8	63.7
			MCDB	61.7	66.9	71.0	73.9	80.6	87.2	88.3	85.2	81.5	76.8	69.6	65.5
2%		WB	54.0	57.5	59.4	64.9	71.1	73.5	76.3	74.6	72.8	69.8	60.1	59.1	
		MCDB	58.0	63.1	67.1	74.2	79.3	83.5	86.2	82.9	80.9	76.1	66.0	61.9	
5%		WB	48.6	52.9	57.2	62.3	69.6	72.4	75.0	73.4	71.2	67.8	56.9	55.0	
		MCDB	54.1	58.6	64.0	71.4	77.3	81.4	84.5	81.2	79.1	72.9	61.1	58.9	
10%		WB	44.8	48.7	54.1	60.0	68.2	71.1	73.7	72.3	69.5	64.2	53.2	50.3	
		MCDB	50.4	55.2	61.8	68.1	75.6	79.0	82.8	79.3	75.6	70.1	58.5	54.0	
Mean Daily Temperature Range	5% DB	MDBR	14.3	16.1	17.0	17.4	14.8	14.2	14.1	13.9	13.7	13.8	14.3	13.0	
		MCDBR	19.2	21.3	20.8	20.0	16.7	15.9	16.1	16.2	17.2	15.4	16.7	17.1	
		MCWBR	15.4	14.7	12.2	10.5	9.3	8.8	7.2	7.2	7.2	8.0	10.5	13.1	
	5% WB	MCDBR	18.2	19.1	19.1	18.7	15.4	14.9	14.5	14.1	14.1	13.4	15.8	15.2	
		MCWBR	16.1	15.7	12.1	11.7	8.9	8.9	7.5	7.3	7.4	8.6	11.9	13.8	
Clear-Sky Solar Irradiance	taub	0.290	0.306	0.339	0.378	0.420	0.466	0.487	0.492	0.394	0.344	0.310	0.293		
	taud	2.528	2.466	2.422	2.332	2.230	2.143	2.111	2.072	2.339	2.473	2.527	2.550		
	Ebn at Noon	287	294	293	285	274	260	254	249	271	276	276	278		
	Edh at Noon	24	29	34	39	44	49	49	50	36	29	24	22		
All-Sky Solar Radiation	RadAvg	718	968	1285	1599	1780	1956	1863	1679	1389	1089	803	629		
	RadStd	56	111	80	123	153	128	107	82	158	124	59	54		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+225	+130

Nomenclature: See separate page